Student ID 1 & 2: *(Do this in pairs)*

Problem Definitions.

Vertex Cover (VC): Given an undirected graph $G = (V, E)$ and an integer k, decide whether there exists a set $S \subseteq V$ with $|S| \leq k$ such that every edge $e \in E$ has at least one endpoint in S.

Set Cover (SC): Given a universe U of n elements, a collection of sets $\mathcal{S} = \{S_1, S_2, \ldots, S_m\}$ where each $S_i \subseteq U$, and an integer k, decide whether there exists a sub-collection $\mathcal{C} \subseteq \mathcal{S}$ with $|\mathcal{C}| \leq k$ such that $\bigcup_{S \in \mathcal{C}} S = U$.

1. **Prove that Set Cover is in NP.**

To show Set Cover $\in$ NP, describe a polynomial-time *verifier*: what is the certificate, and how does the verifier check it?

2. **Prove that Set Cover is NP-Hard.**

You may assume Vertex Cover is NP-Complete. Show a polynomial-time reduction from Vertex Cover to Set Cover, i.e., show VC $\leq_p$ SC.

Your answer should include:

(a) The construction: given a VC instance (G, k), define the corresponding SC instance $(U, \mathcal{S}, k')$.

(b) Correctness (forward direction): if (G, k) is a YES instance of VC, then $(U, \mathcal{S}, k')$ is a YES instance of SC.

(c) Correctness (backward direction): if $(U, \mathcal{S}, k')$ is a YES instance of SC, then (G, k) is a YES instance of VC.

(d) Argue that the construction runs in polynomial time.